



Chapter 9

JavaScript: Functions

Internet & World Wide Web
How to Program, 5/e

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OBJECTIVES

In this chapter you will:

- Construct programs modularly from small pieces called functions.
- Define new functions.
- Pass information between functions.
- Use simulation techniques based on random number generation.
- Use the new HTML5 **audio** and **video** elements
- Use additional global methods.
- See how the visibility of identifiers is limited to specific regions of programs.

9.1 Introduction

- ▶ To develop and maintain a **large program**

演算法概念:

- construct it from **small, simple** pieces
- **divide** and **conquer** (將大問題切割成小問題，再處理)
- 註：大程式由小程式或簡單模組組成，較有結構性和系統性

- ▶ You can combine your functions with **prepackaged functions** (內建函式) and **objects** (內建物件) in JavaScript

- ▶ The prepackaged functions (such as **Math.pow**) are called **methods** (物件之方法).

 *課外補充 : [Math物件\[連結\]](#)

- ▶ JavaScript provides **several objects** such as



- ▶ **mathematical calculations** (數學計算)
- ▶ **string manipulations** (字串處理)
- ▶ **date and time manipulations** (時間/日期控制),etc.

註：JavaScript針對數學算式、字串處理、時間/日期控制有提供相關物件。



9.2 Program Modules in JavaScript (Cont.)

- ▶ You can define **programmer-defined functions** (自訂函式) that perform **specific tasks** (特定工作) and use them at many points in a script
 - 註：可自訂函式，執行特定工作，以廣泛使用

函式名稱(參數1, 參數2, ...); //函式呼叫方式

- ▶ **Functions** are invoked by its **function name**, followed by a left parenthesis “(”, followed by a **comma-separated list of zero or more arguments** (0或多個參數), followed by a right parenthesis “)”.
 - 註：呼叫函式時，透過呼叫函式的名稱再加上0~多個參數。

物件.方法名稱 (參數1, 參數2, ...); //物件方法 呼叫方式(ex: **Math.pow(x,y);**)

- ▶ **Methods** are called in the same way, but require the **name of the object** (需要物件名稱) to which the method belongs and a **dot “.”** preceding the method name
 - 註：呼叫方法時，需要有物件名稱，再加上“點”，ex: **Math.pow(x,y);**

參數

- ▶ May be **constants** (常數), **variables** (變數) or **expressions** (算式).
 - 註：參數可以是數字(ex: 1, 2)、變數(ex: x, y)、或算式(ex: 2x+3y)。

return //回傳

- **passes information** from a function **back to** the point where it was called
- 註：將function內部資料傳送回呼叫的地方

9.3.1 Programmer-Defined Function **square**

函式定義

- ▶ The format of a function definition is

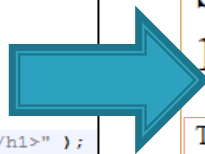
function 函式名稱 (參數列)

{

declarations and statements (宣告、動作敘述)

}

```
1) SquareInt.html x
1 <!DOCTYPE html>
2
3 <!-- Fig. 9.2: SquareInt.html -->
4 <!-- Programmer-defined function square. -->
5 <html>
6 <head>
7 <meta charset = "utf-8">
8 <title>A Programmer-Defined square Function</title>
9 <style type = "text/css">
10 p { margin: 0; }
11 </style>
12 <script>
13
14 document.writeln( "<h1>Square the numbers from 1 to 10</h1>" );
15
16 // square the numbers from 1 to 10
17 for ( var x = 1; x <= 10; ++x )
18 document.writeln( "<p>The square of " + x + " is " +
19 square( x ) + "</p>" );
20
21 // The following square function definition body is executed
22 // only when the function is called explicitly as in line 19
23 function square( y )
24 {
25 return y * y;
26 } // end function square
27
28 </script>
29 </head><body></body>
30 </html>
```



A Programmer-Defined x

file:///D:/D槽/Dropbox/105下(網路程式 ☆

Square the numbers from 1 to 10

(平方結果)

The square of 1 is 1
The square of 2 is 4
The square of 3 is 9
The square of 4 is 16
The square of 5 is 25
The square of 6 is 36
The square of 7 is 49
The square of 8 is 64
The square of 9 is 81
The square of 10 is 100

9.3.1 Programmer-Defined Function **square** (cont.)

函式結束 (三種方式)

▶ Three ways back to the point where a function was invoked

- 1. 函式結束的括弧"}"
- 2. 執行 **return**; (ex: return y*y;)
- 3. 執行 **return** 算式;

```
2) maximum.html x
1  <!DOCTYPE html>
2
3  <!-- Fig. 9.3: maximum.html -->
4  <!-- Programmer-Defined maximum function. -->
5  <html>
6  <head>
7    <meta charset = "utf-8">
8    <title>Maximum of Three Values</title>
9    <style type = "text/css">
10     p { margin: 0; }
11    </style>
12    <script>
13
14     var input1 = window.prompt( "Enter first number", "0" );
15     var input2 = window.prompt( "Enter second number", "0" );
16     var input3 = window.prompt( "Enter third number", "0" );
17
18     var value1 = parseFloat( input1 );
19     var value2 = parseFloat( input2 );
20     var value3 = parseFloat( input3 );
21
22     var maxValue = maximum( value1, value2, value3 );
23
24     document.writeln( "<p>First number: " + value1 + "</p>" +
25                       "<p>Second number: " + value2 + "</p>" +
26                       "<p>Third number: " + value3 + "</p>" +
27                       "<p>Maximum is: " + maxValue + "</p>" ); (輸出)
28
29     // maximum function definition (called from line 22)
30     function maximum( x, y, z )
31     {
32       return Math.max( x, Math.max( y, z ) );
33     } // end function maximum (函式定義)
34
35    </script>
36  </head><body></body>
37 </html>
```

這個網頁顯示：

Enter first number

這個網頁顯示：

Enter second number

☐ 防止此網頁產生其他對話方塊。

這個網頁顯示：

Enter third number

☐ 防止此網頁產生其他對話方塊。

Maximum of x

file:///D:/D槽/Dropbox/10

First number: 10
Second number: 20
Third number: 30
Maximum is: 30

9.4 Notes on Programmer-Defined Functions

- ▶ All variables declared with the keyword **var** in **function** definitions are **local variables** (區域變數)
 - This means that they can be accessed **only in** the function.
 - 註：函式內宣告的變數為區域變數，只能在函式內使用。
- ▶ A **function's parameters** (函式的參數) are also considered to be local variables.
 - 註：函式的參數也是區域變數

函式的優點

- ▶ If a program is **separated** (分割) **by** functions
 - Makes program development **more manageable** (易於管理).
 - Software **reusability** (可重複利用).
 - **Avoid repeating code** in a program (避免寫重複的程式碼).

9.5 Random Number Generation

Math.random(...); //產生隨機數

- ▶ generates a **floating-point** value from $0.0 \leq \text{隨機數} < 1.0$ (不等於1.0)
- ▶ To **random an integer** in a **certain range**, we can use **scaling** (縮放) and **shifting** (平移) the values, and then using **Math.floor** (取下高斯) to convert them to integers

- **scaling factor** determines the **range**
 - 註：**Scaling**決定數字範圍
- **shift number** determines the **range begins**
 - 註：**shifting**決定啟始數字
- **Math.floor** makes it to the closest **integer**
 - 註：透過下高斯，轉換成整數
(問題：如何產生1~6 隨機數?)
 - 範例：**value = Math.floor(1 + Math.random() * 6);**

Math.random(...); //產生隨機數 (通用法)

- ▶ We can **generalize** this result as
face = Math.floor(a + Math.random() * b);
 - **a**: is the **shifting value** (平移係數) (隨機數的最小值)
 - **b**: is the **scaling factor** (放大係數) (隨機數的範圍).

(產生1~6的隨機數)

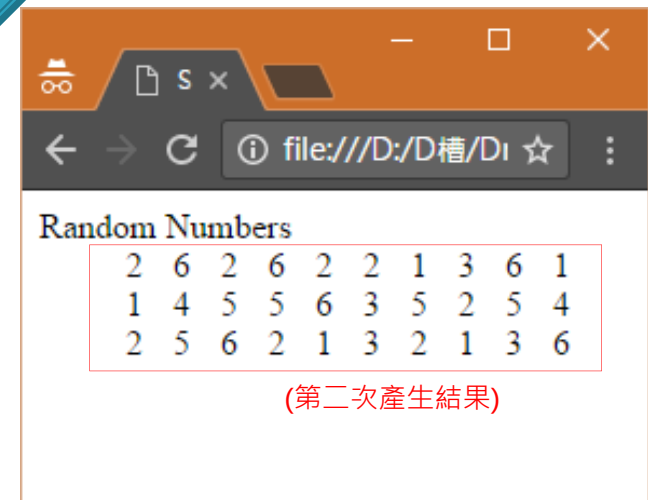
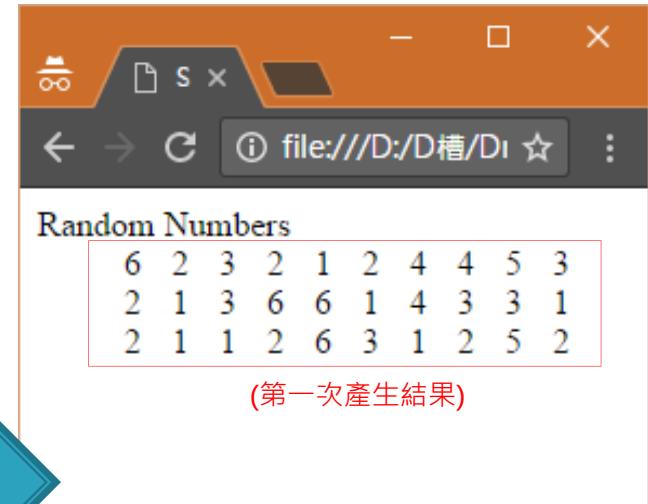
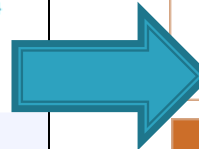
(平移)

(範圍放大至 $0.0 \leq \dots < 6.0$)

9.5 Random Number Generation

- ▶ Example: 隨機產生30個1~6的數字 (看範例)

```
3) RandomInt.html x
1  <!DOCTYPE html>
2
3  <!-- Fig. 9.4: RandomInt.html -->
4  <!-- Random integers, shifting and scaling. -->
5  <html>
6  <head>
7      <meta charset = "utf-8">
8      <title>Shifted and Scaled Random Integers</title>
9      <style type = "text/css">
10         p, ol { margin: 0; }
11         li { display: inline; margin-right: 10px; }
12      </style>
13      <script>
14
15         var value;
16
17         document.writeln( "<p>Random Numbers</p><ol>" );
18
19         for ( var i = 1; i <= 30; ++i )
20         {
21             value = Math.floor( 1 + Math.random() * 6 );
22             document.writeln( "<li>" + value + "</li>" );
23         } // end for
24
25         document.writeln( "</ol>" );
26
27      </script>
28  </head><body></body>
29  </html>
```



9.5.2 Displaying Random Images

(可以玩玩看，範例程式)

- ▶ Example:  [\[連結\]](#) Random image generator
 - To displays **four** randomly selected die images (4個骰子圖) every time the user **clicks** a “Roll Dice button(擲骰子)” on the page.

```
4) RollDice.html
<html>
<head>
  <meta charset = "utf-8">
  <title>Random Dice Images</title>
  <style type = "text/css">
    li { display: inline; margin-right: 10px; }
    ul { margin: 0; }
  </style>
  <script>
    // variables used to interact with the img elements
    var die1Image;
    var die2Image;
    var die3Image;
    var die4Image;
    (2) start函式: 取得id物件
    function start()
    {
      (取得 id = rollButton元件)
      var button = document.getElementById( "rollButton" );
      button.addEventListener( "click", rollDice, false );
      die1Image = document.getElementById( "die1" );
      die2Image = document.getElementById( "die2" );
      die3Image = document.getElementById( "die3" );
      die4Image = document.getElementById( "die4" );
      (3) 註冊 eventhandler: 當click時執行 rollDice函式)
    } // end function start
    (4) rollDice函式: 更新骰子圖片
    function rollDice()
    {
      setImage( die1Image );
      setImage( die2Image );
      setImage( die3Image );
      setImage( die4Image );
    } // end function rollDice
```

```
(5) setImage函式: 寫入骰子圖片屬性
34 // set src and alt attributes for a die
35 function setImage( dieImg )
36 {
37   (6) 隨機產生骰子點數性
38   var dieValue = Math.floor( 1 + Math.random() * 6 );
39   dieImg.setAttribute( "src", "die" + dieValue + ".png" );
40   (7) 寫入骰子圖片屬性
41   dieImg.setAttribute( "alt", "die image with " + dieValue + " spot(s)" );
42 } // end function setImage
43 window.addEventListener( "load", start, false );
44 </script>
45 </head>
46 <body>
47   <form action = "#">
48     <input id = "rollButton" type = "button" value = "Roll Dice">
49   </form>
50   ( id = rollButton元件)
51   <ol>
52     <li><img id = "die1" src = "blank.png" alt = "blank image"></li>
53     <li><img id = "die2" src = "blank.png" alt = "blank image"></li>
54     <li><img id = "die3" src = "blank.png" alt = "blank image"></li>
55     <li><img id = "die4" src = "blank.png" alt = "blank image"></li>
56   </ol>
57 </body>
58 </html>
```

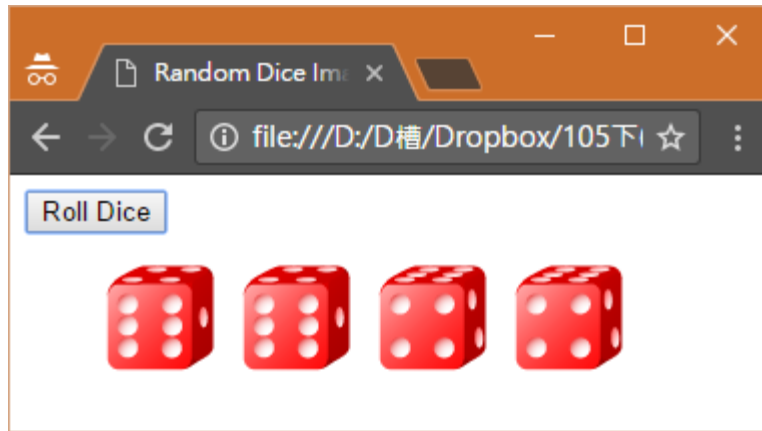
***補充: addEventListener()**

(*總結) 流程: 取得id→註冊函式→按鈕觸發→更換圖片src

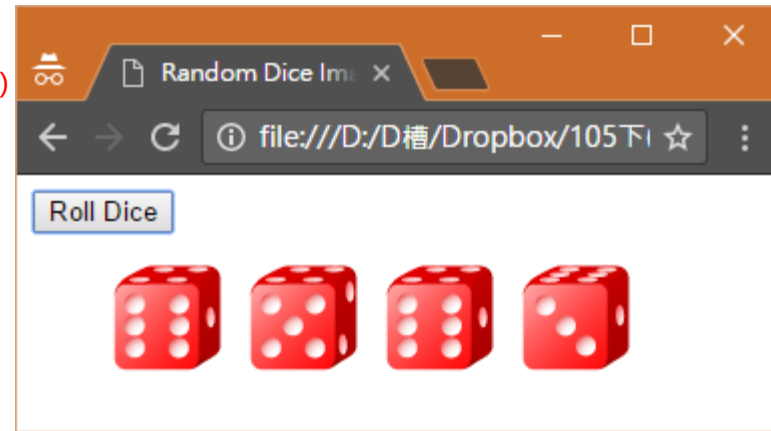
9.5.2 Displaying Random Images

▶ Example:  [\[連結\]](#) Random image generator

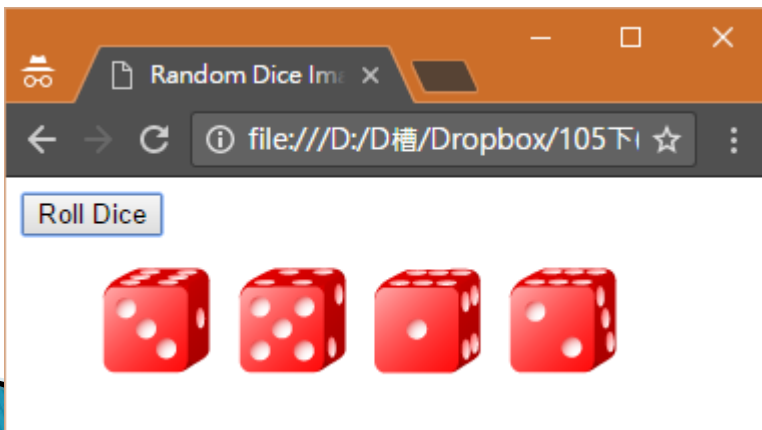
(第一次產生)



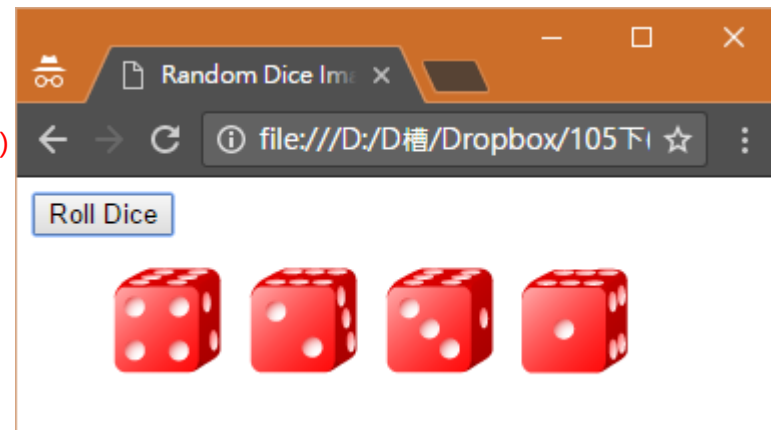
(第三次產生)



(第二次產生)



(第四次產生)



9.5.2 Displaying Random Images

(類似視窗語言)

User Interactions (使用者互動) Via Event Handling (事件處理)

- ▶ Previously, we **interact** with users by a **prompt dialog** (對話框) or an **alert dialog** (警示框).
 - ▶ 註：先前**固定**使用對話框和警示框和使用者互動：**不具彈性**
- ▶ A **prompt dialog** (對話框) can obtain only **one value** at a time
 - ▶ 註：使用對話框和一次只能獲得一項資訊。

GUI event handling 視覺化事件處理

- ▶ **Inputs (輸入)** are received from the user via an HTML5 **form** (表單).
 - ▶ 註：通常透過表單(form)取得使用者輸入(input)
- ▶ **Outputs (輸出)** are displayed to the user in the web page.
 - ▶ 註：通常輸出呈現在網頁。
- ▶ Interact with user's by **an element** in a form.
 - ▶ 註：透過表單互動產生事件(ex: 按下"按鈕").

9.5.2 Displaying Random Images

`<form action = "#"> ... </form>` 用於表單方式跟使用者互動

- ▶ It requires a **form** with an **action** attribute.
 - ▶ This form **does not post** its information to a web server, the string **"#"** is used to allow this document to validate.
 - ▶ 註：透過**form**的型式觸發，但實際沒有將資訊送出，因此以**action="#"**暫替。
 - ▶ 註：**"#"**：代表**目前頁面**的意思。

`<input id = "rollButton" type = "button" value = "...">` 透過表單"按鈕"觸發事件

- ▶ To interact with an **element** in the web page, the script is **notified of (通知)** the event and then **processes the event (處理事件)**.
 - ▶ 註：經由和網頁的**互動(動作)**，script會接收這個通知並處理此事件

*//轉換成 **button** 物件*
`button = document.getElementById( "rollButton" );`
`button.addEventListener( "click", rollDice, false );`

- ▶ The **button click (按"按鈕")** is known as the **event** and will **be processed** by the **event handler (事件處理器)**.
 - ▶ Then, a specified **event-handling function** will be called.
 - ▶ 註：透過使用者**按下按鈕"click"**來觸發事件，並經由**event-handler**來呼叫對應的**rollDice()函式**。
 - ▶ 註：第三個欄位填 **false** (預設值)。

9.5.2 Displaying Random Images

**

透過修改img的src屬性，達成更換圖片

▶ The four **img** elements will display **randomly**.

▪ 註：4張骰子圖隨機呈現

▶ Their **id** attributes (**die1**, **die2**, **die3** and **die4**, respectively) must have a **unique value**

▪ 註：定義的id值需唯一（不能重複），以便指定變更內容。

▶ When the page **first renders**, each **img element** displays the image **blank.png** (an empty white image).

▪ 註：第一次預覽時，會呈現blank.png（空白圖）。

▪ 註：最後，透過**setImage()**來修改圖片屬性(ex: **src**及**alt**等)

```
function setImage( dieImg )  
{  
    var dieValue = Math.floor( 1 + Math.random() * 6 );  
    dieImg.setAttribute( "src", "die" + dieValue + ".png" );  
    dieImg.setAttribute( "alt",  
        "die image with " + dieValue + " spot(s)" );  
} // end function setImage
```

```
die1Image = document.getElementById( "die1" );  
die2Image = document.getElementById( "die2" );  
die3Image = document.getElementById( "die3" );  
die4Image = document.getElementById( "die4" );
```


9.5.2 Displaying Random Images

//註冊(建立關聯/註冊: 事件&函式)

window.addEventListener("load", start, false);

瀏覽器讀取網頁時，呼叫“start()”函式

▶ To specify the **function** to call when an **event** occurs, you can register an **event handler** for that event.

■ 註：向 **event handler** 將 **event** 關聯至一個 **function**

▶ Method **addEventListener** takes three arguments:

- ▶ The first: **event name** (事件名稱) for which you want to register to a handler
* 課外補充：[addEventListener\(\)](#)
 - ▶ 註：第一欄填“事件名稱”，如：“load”、“click”等
- ▶ The second: the function name (to be called to handle the event)
 - ▶ 註：第二欄填“欲呼叫的函式名稱”，如：“start”、“rollDice”等
- ▶ The third: typically false—the true value is beyond this book’s scope
 - ▶ 註：第三欄填“false” (不在本書討論範圍)

9.5.3 Rolling Dice Repeatedly and Displaying Statistics



(可以玩玩看，範例程式)

- ▶ Example: To roll 12 dice at a time and keep statistics showing the number of times. [\[連結\]擲骰子及建表統計](#)

```
5) RollDice2.html
1 <!DOCTYPE html>
2
3 <!-- Fig. 9.6: RollDice.html -->
4 <!-- Rolling 12 dice and displaying frequencies. -->
5 <html>
6 <head>
7 <meta charset = "utf-8">
8 <title>Die Rolling Frequencies</title>
9 <style type = "text/css">
10     img { margin-right: 10px; }
11     table { width: 200px;
12             border-collapse: collapse;
13             background-color: lightblue; }
14     table, td, th { border: 1px solid black;
15                     padding: 4px;
16                     margin-top: 20px; }
17     th { text-align: left;
18          color: white;
19          background-color: darkblue; }
20 </style>
21 <script>
22     var frequency1 = 0;
23     var frequency2 = 0;
24     var frequency3 = 0;
25     var frequency4 = 0;
26     var frequency5 = 0;
27     var frequency6 = 0;
28     var totalDice = 0;
29
30     // register button event handler
31     function start()
32     {
33         var button = document.getElementById( "rollButton" );
34         button.addEventListener( "click", rollDice, false );
35     } // end function start
36
37     // roll the dice
38     function rollDice()
39     {
40         var face; // face rolled
41     }
```

(2) start函式

(取得 id = rollButton 元件，轉成物件來監聽)

(3) 註冊 eventhandler：當click時執行rollDice函式

(4) rollDice函式

```
42 // loop to roll die 12 times
43 for ( var i = 1; i <= 12; ++i )
44 {
45     face = Math.floor( 1 + Math.random() * 6 );
46     tallyRolls( face ); // increment a frequency counter
47     setImage( i, face ); // display appropriate die image
48     ++totalDice; // increment total
49 } // end die rolling loop
50
51 updateFrequencyTable();
52 } // end function rollDice
53
54 // increment appropriate frequency counter
55 function tallyRolls( face )
56 {
57     switch ( face )
58     {
59         case 1:
60             ++frequency1;
61             break;
62         case 2:
63             ++frequency2;
64             break;
65         case 3:
66             ++frequency3;
67             break;
68         case 4:
69             ++frequency4;
70             break;
71         case 5:
72             ++frequency5;
73             break;
74         case 6:
75             ++frequency6;
76             break;
77     } // end switch
78 } // end function tallyRolls
79
80 // set image source for a die
81 function setImage( dieNumber, face )
82 {
```

(5) tallyRolls函式：統計骰子次數

(6) setImage函式：更換骰子圖片

9.5.3 Rolling Dice Repeatedly and Displaying Statistics

- ▶ Example: To roll 12 dice at a time and keep statistics showing the number of times.

(id轉成物件，以便修改屬性)

//12張圖

(7) updateFrequencyTable函式: 更新統計Table欄位

(6) 取得Table物件

(7) 寫入Table內容

(8) formFrequency函式

(1) 註冊event handler: 當load時，執行start函式

```
123 <p><img id = "die1" src = "blank.png" alt = "die 1 image">
124 <img id = "die2" src = "blank.png" alt = "die 2 image">
125 <img id = "die3" src = "blank.png" alt = "die 3 image">
126 <img id = "die4" src = "blank.png" alt = "die 4 image">
127 <img id = "die5" src = "blank.png" alt = "die 5 image">
128 <img id = "die6" src = "blank.png" alt = "die 6 image"></p>
129 <p><img id = "die7" src = "blank.png" alt = "die 7 image">
130 <img id = "die8" src = "blank.png" alt = "die 8 image">
131 <img id = "die9" src = "blank.png" alt = "die 9 image">
132 <img id = "die10" src = "blank.png" alt = "die 10 image">
133 <img id = "die11" src = "blank.png" alt = "die 11 image">
134 <img id = "die12" src = "blank.png" alt = "die 12 image"></p>
135 <form action = "#">
136 <input id = "rollButton" type = "button" value = "Roll Dice">
137 </form>
138 <div id = "frequencyTableDiv"></div>
139 </body>
140 </html>
```

//空id:之後要插入html table語法

```
141 <!--
142 *****
143 * (C) Copyright 1992-2012 by Deitel & Associates, Inc. and
144 * Pearson Education, Inc. All Rights Reserved.
145 *
146 * DISCLAIMER: The authors and publisher of this book have used their
147 * best efforts in preparing the book. These efforts include the
148 * development, research, and testing of the theories and programs
149 * to determine their effectiveness. The authors and publisher make
150 * no warranty of any kind, expressed or implied, with regard to these
151 * programs or to the documentation contained in these books. The authors
152 * and publisher shall not be liable in any event for incidental or
153 * consequential damages in connection with, or arising out of, the
154 * furnishing, performance, or use of these programs.
155 *****
156
```

length: 6,5 Ln: 1 Col: 1 Sel: 0|0

Windows (CR LF) UTF-8

INS

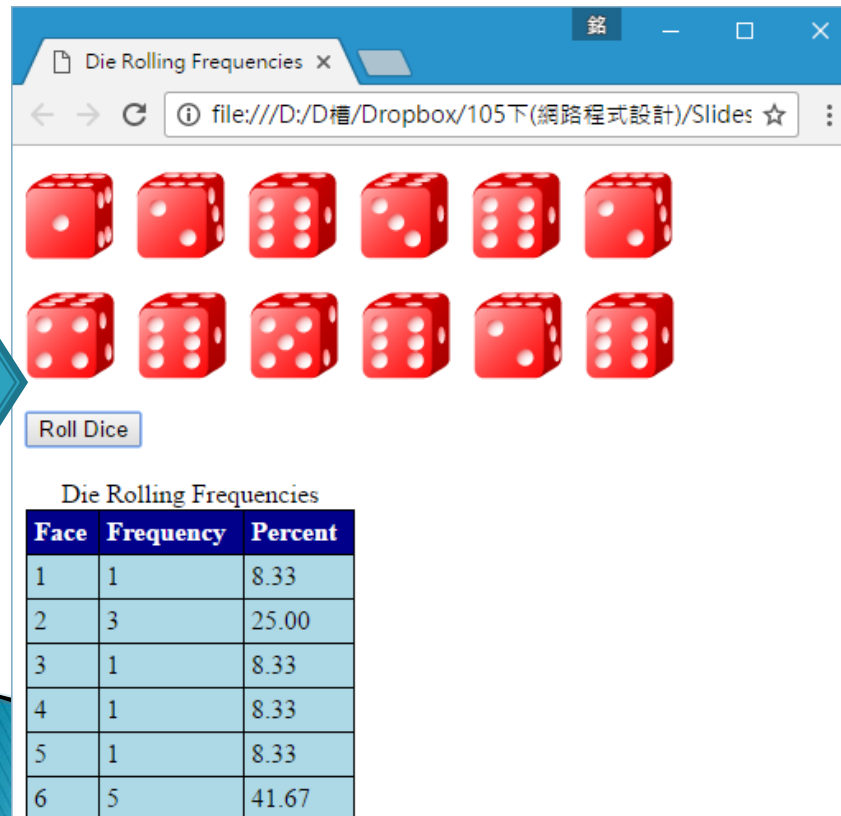
9.5.3 Rolling Dice Repeatedly and Displaying Statistics

- ▶ Example: To roll 12 dice at a time and keep statistics showing the number of times.

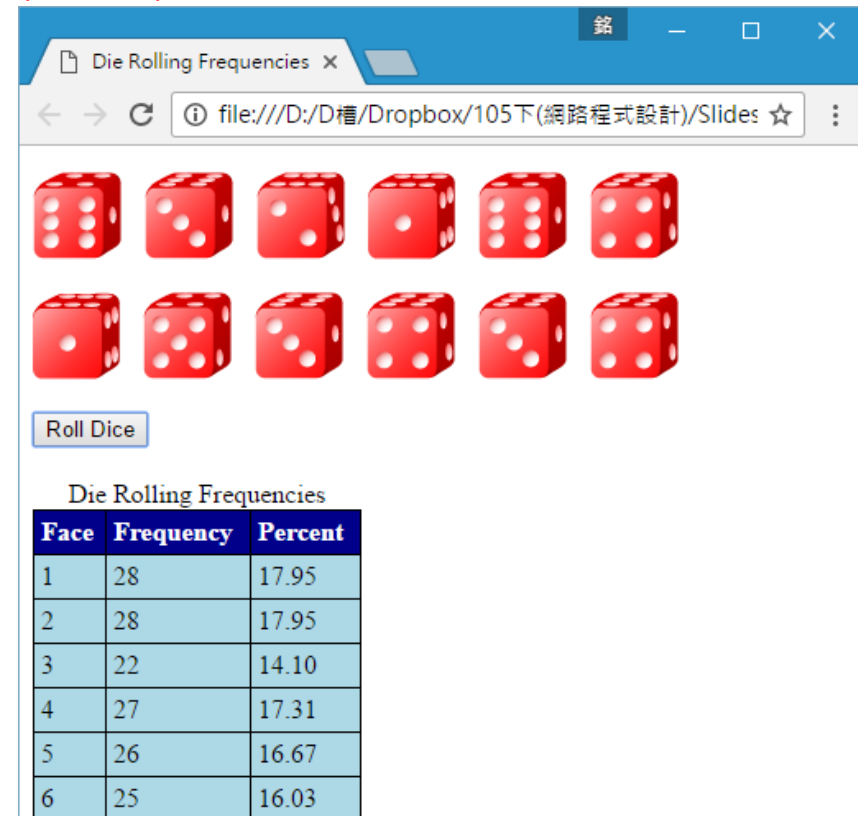


[連結]擲骰子及建表統計

(第一次產生)

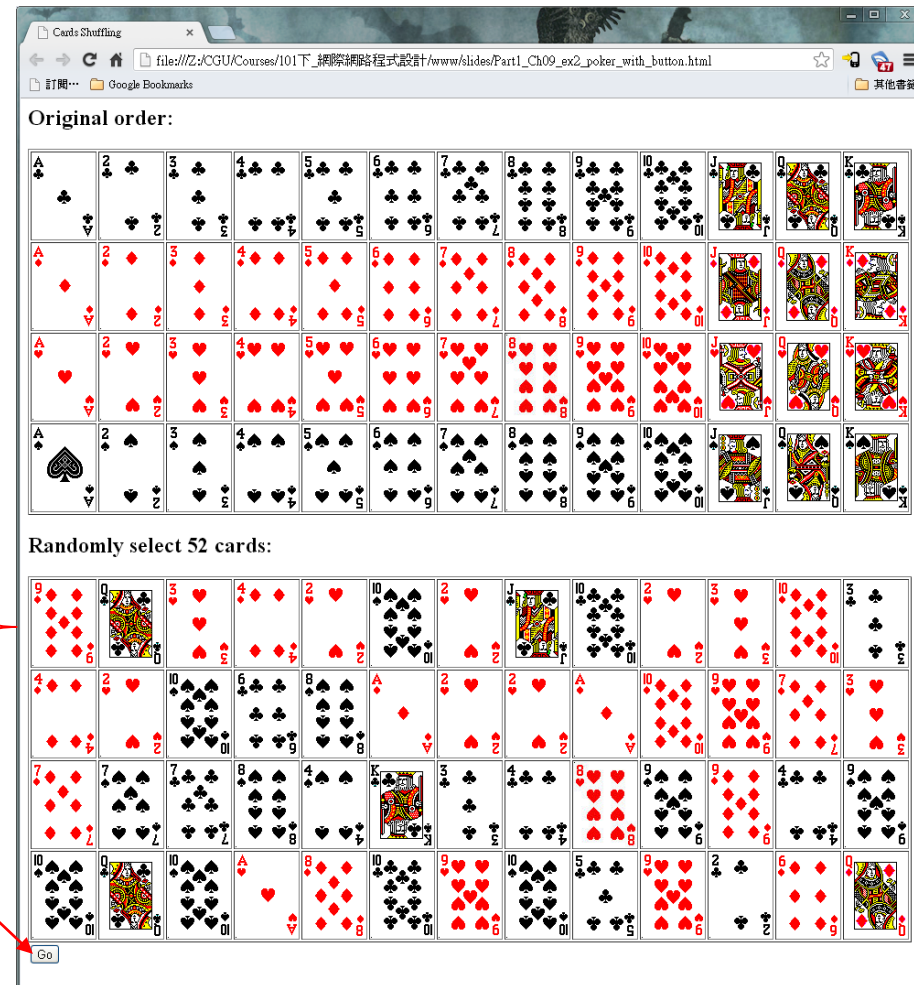


(第x次產生)



In-Class Exercise #1

- ▶ 課中練習：
- ▶ 請利用JavaScript產生右方表格
 - 1. 上表依序列出52張牌
 - 2. 當開啟網頁時，下表隨機挑選上方一張牌
 - 允許重複挑選
 - 使用load事件
 - 3. 最下方加入一個按鈕來觸發重新挑選
 - 允許重複出現
 - 使用click事件
 - 4. 發完牌後..



(圖片檔案：在ftp的.../porkers資料匣)

(最後留時間給同學們練習)



9.6 Example: Game of Chance; Introducing the HTML5 audio and video Elements

▶ Example:  [連結] The game of craps. (程式碼約: 200行，小複雜)

```
6) Craps.html
1 <!DOCTYPE html>
2
3 <!-- Fig. 9.7: Craps.html -->
4 <!-- Craps game simulation. -->
5
6 <html>
7   <head>
8     <meta charset = "utf-8">
9     <title>Craps Game Simulation</title>
10    <style type = "text/css">
11      p.red { color: red }
12      img { width: 54px; height: 54px; }
13      div { border: 5px ridge royalblue;
14            padding: 10px; width: 120px;
15            margin-bottom: 10px; }
16      .point { margin: 0px; }
17    </style>
18    <script>
19      // variables used to refer to page elements
20      var pointDie1Img; // refers to first die point img
21      var pointDie2Img; // refers to second die point img
22      var rollDie1Img; // refers to first die roll img
23      var rollDie2Img; // refers to second die roll img
24      var messages; // refers to "messages" paragraph
25      var playButton; // refers to Play button
26      var rollButton; // refers to Roll button
27      var dicerolling; // refers to audio clip for dice
28
29      // other variables used in program
30      var myPoint; // point if no win/loss on first roll
31      var die1Value; // value of first die in current roll
32      var die2Value; // value of second die in current roll
33
34      (4) startGame函式:
35      // starts a new game
36      function startGame()
37      {
38        // (初始化：將所需元件，轉成物件)
39        // get the page elements that we'll interact with
40        dicerolling = document.getElementById( "dicerolling" );
41        pointDie1Img = document.getElementById( "pointDie1" );
42        pointDie2Img = document.getElementById( "pointDie2" );
43        rollDie1Img = document.getElementById( "rollDie1" );
44        rollDie2Img = document.getElementById( "rollDie2" );
45        messages = document.getElementById( "messages" );
46        playButton = document.getElementById( "play" );
47        rollButton = document.getElementById( "roll" );
```

```
45
46 // prepare the GUI (5) 不允許按此按鈕 (變灰色)
47 rollButton.disabled = true; // disable rollButton
48 setImage( pointDie1Img ); // reset image for new game
49 setImage( pointDie2Img ); // reset image for new game
50 setImage( rollDie1Img ); // reset image for new game
51 setImage( rollDie2Img ); // reset image for new game
52
53 myPoint = 0; // there is currently no point
54 firstRoll(); // roll the dice to start the game
55 } // end function startGame
56
57 (6) firstRoll 函式: 第一輪數字判定
58 // perform first roll of the game
59 function firstRoll()
60 {
61   var sumOfDice = rollDice(); // first roll of the dice
62   // (第一輪數字7, 11 則贏; 若為2, 3, 12 則輸; 其他, 記錄下來)
63   // determine if the user won, lost or must continue rolling
64   switch (sumOfDice)
65   {
66     case 7: case 11: // win on first roll //寫入html訊息
67       messages.innerHTML =
68         "You Win!!! Click Play to play again.";
69       break;
70     case 2: case 3: case 12: // lose on first roll
71       messages.innerHTML =
72         "Sorry. You Lose. Click Play to play again.";
73       break;
74     default: // remember point
75       myPoint = sumOfDice; //若是其他：記錄目前點數值
76       setImage( pointDie1Img, die1Value );
77       setImage( pointDie2Img, die2Value );
78       messages.innerHTML = "Roll Again!";
79       rollButton.disabled = false; // enable rollButton
80       playButton.disabled = true; // disable playButton
81       break;
82   } // end switch
83 } // end function firstRoll
84
85 (8) rollAgain函式: 再骰一次
86 // called for subsequent rolls of the dice
87 function rollAgain()
88 {
89   var sumOfDice = rollDice(); // subsequent roll of the dice
90   if (sumOfDice == myPoint)
91   {
```

9.6 Example: Game of Chance; Introducing the HTML5 audio and video Elements



▶ Example: the game of craps.

(3) 取得物件id · 註冊event-handler: play click->startgame()、roll click-> rollAgain()、聲音ended->showDice

(2) start函式: 註冊event-handler

```
91 messages.innerHTML =
92 "You Win!!! Click Play to play again.";
93 rollButton.disabled = true; // disable rollButton
94 playButton.disabled = false; // enable playButton
95 } // end if
96 else if (sumOfDice == 7) // craps
97 {
98     messages.innerHTML =
99     "Sorry. You Lose. Click Play to play again.";
100     rollButton.disabled = true; // disable rollButton
101     playButton.disabled = false; // enable playButton
102 } // end else if
103 } // end function rollAgain
104
105 (9) rollDice函式: 隨機產生點數
106 // roll the dice
107 function rollDice()
108 {
109     dicerolling.play(); // play dice rolling sound
110     //撥放audio (音效檔)
111     // clear old die images while rolling sound plays
112     die1Value = NaN;
113     die2Value = NaN;
114     showDice();
115     (更正程式碼)
116     die1Value = Math.floor(1 + Math.random() * 6);
117     die2Value = Math.floor(1 + Math.random() * 6);
118     return die1Value + die2Value;
119 } // end function rollDice
120
121 (10) showDice函式: 寫入骰子圖片
122 // display rolled dice
123 function showDice()
124 {
125     setImage( rollDie1Img, die1Value );
126     setImage( rollDie2Img, die2Value );
127 } // end function showDice
128
129 (5) setImage函式: 寫入圖片來源src
130 // set image source for a die
131 function setImage( dieImg, dieValue )
132 {
133     if ( isFinite( dieValue ) )
134         dieImg.src = "die" + dieValue + ".png";
135     else //透過 .屬性 (也可以)
136         dieImg.src = "blank.png";
137 } // end function setImage
138
139 // register event listeners
```

```
136 // register event listeners
137 function start()
138 {
139     var playButton = document.getElementById( "play" );
140     playButton.addEventListener( "click", startGame, false );
141     var rollButton = document.getElementById( "roll" );
142     rollButton.addEventListener( "click", rollAgain, false );
143     var diceSound = document.getElementById( "dicerolling" );
144     diceSound.addEventListener( "ended", showDice, false );
145 } // end function start
146
147 window.addEventListener( "load", start, false );
148
149 </script>
150 </head>
151 <body>
152     <audio id = "dicerolling" preload = "auto">
153         <source src = "http://test.deitel.com/dicerolling.mp3"
154             type = "audio/mpeg">
155         <source src = "http://test.deitel.com/dicerolling.ogg"
156             type = "audio/ogg">
157     </audio>
158     Browser does not support audio tag</audio>
159     <p><a href = "CrapsRules.html">Click here for a short video
160     explaining the basic Craps rules</a></p>
161     <div id = "pointDiv">
162         <p class = "point">Point is:</p>
163         <img id = "pointDie1" src = "blank.png"
164             alt = "Die 1 of Point Value">
165         <img id = "pointDie2" src = "blank.png"
166             alt = "Die 2 of Point Value">
167     </div>
168     <div class = "rollDiv">
169         <img id = "rollDie1" src = "blank.png"
170             alt = "Die 1 of Roll Value">
171         <img id = "rollDie2" src = "blank.png"
172             alt = "Die 2 of Roll Value">
173     </div>
174     <form action = "#">
175         <input id = "play" type = "button" value = "Play">
176         <input id = "roll" type = "button" value = "Roll" disabled>
177     </form>
178     <p id = "messages" class = "red">Click Play to start the game</p>
179 </body>
180 </html>
```

(★javascript 可開發複雜遊戲)

9.6 Example: Game of Chance; Introducing the HTML5 audio and video Elements

- ▶ Example:
- ▶ [\[連結\] The game of craps](#)

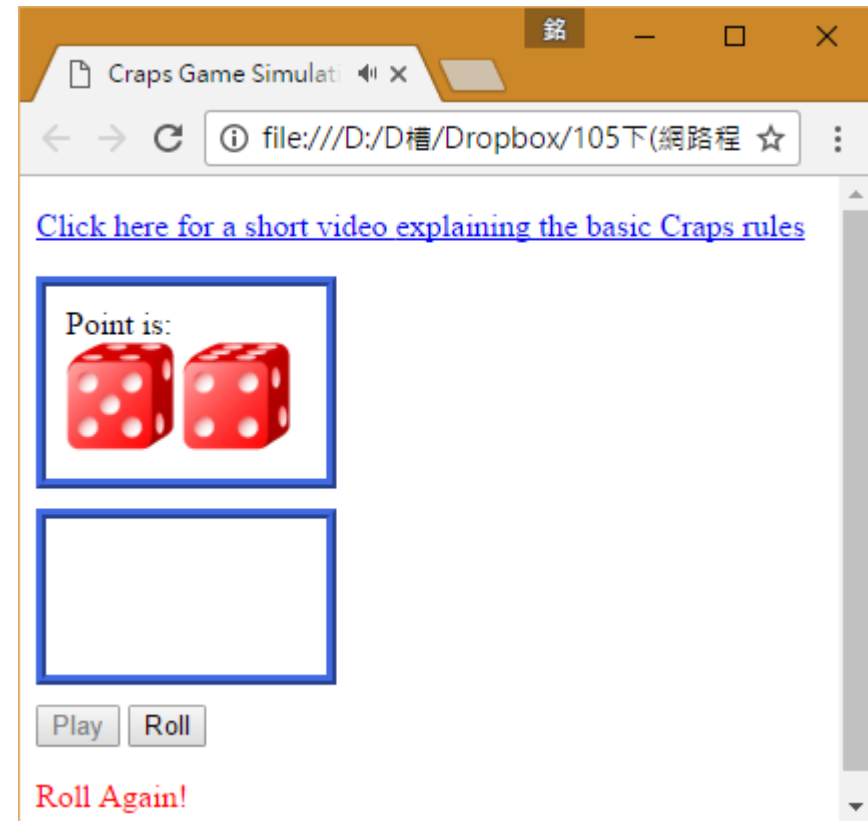
(初始)

(遊戲規則：

第一輪：若數字加總為 7或11，則贏；

若數字加總為 2或3或12，則輸；

若其他加總數字，則記錄下來，進入第二輪)



9.6 Example: Game of Chance; Introducing the HTML5 audio and video Elements

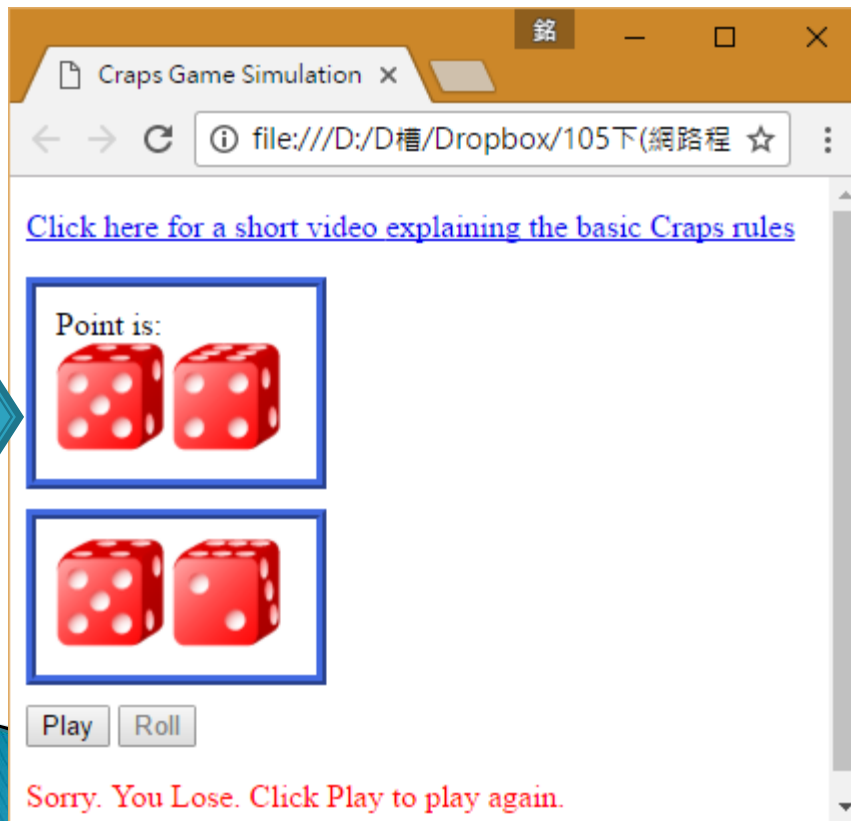
▶ Example: [\[連結\] The game of craps](#)

(第二輪:

若數字加總跟上一輪相同，則贏;

若加總為7，則輸; 其他加總數字，則繼續)

(示意：點數加總跟上一輪相同，則贏)



9.6 Example: Game of Chance; Introducing the HTML5 **audio** and **video** Elements (Cont.)



<audio id = "... " preload = "auto">

audio 元件

- ▶ With specified **id**, we can **programmatically control** the audio to **play** (撥放, **.play();**) and **handle** the **audio event**.
 - 註：透過**取得audio元件id**，則可以**程式控制播放**或**抓取播放事件**
- ▶ **"auto"**: to download the audio in advanced.
 - 註：**auto**: **預先下載** (適用網路較慢環境).
 - 註：大部分瀏覽器支援**MP3**、**OGG** or **WAV**格式。
(註：**OGG**:開放標準的多媒體檔案格)

<source src = "... " type = "audio/ogg">

- ▶ Each source element specifies a **src** (來源) and a **type** (類型/檔案格式) attribute.
 - 註：**src** 描述**audio來源位址**，**type**描述**audio格式** (ex: mp3, wav, etc.)
 - 註：當**src**有多個位址時，以**第一個位址**代表。

- (★補充) 線上音效網站：

<https://www.findsounds.com/index.html>

9.6 Example: Game of Chance; Introducing the HTML5 audio and video Elements (Cont.)



[\[連結\] The game of craps](#)

<video controls>

<source src = "... " type = "... ">

</video>

Video元件

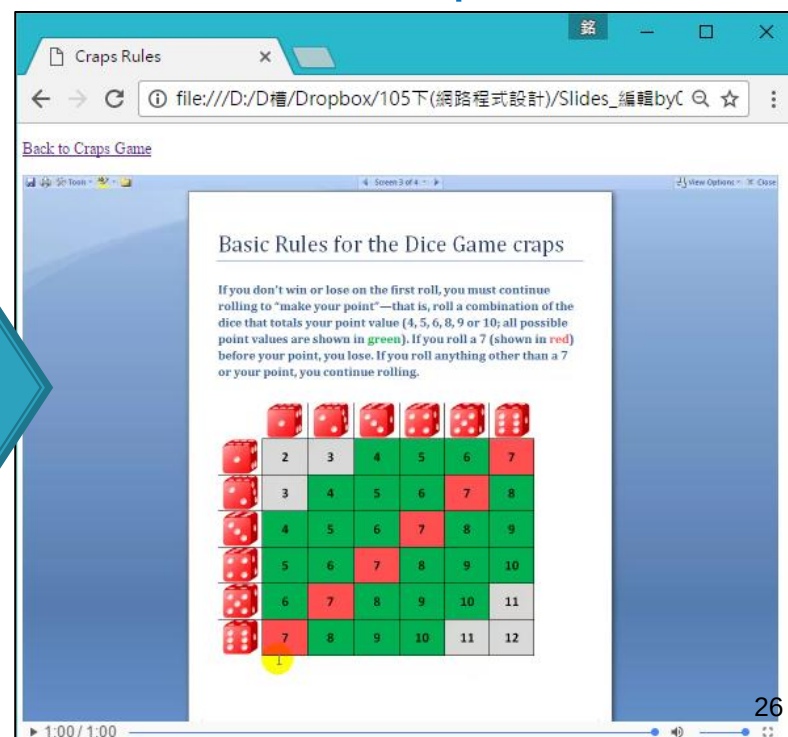
(★補充：WebM是一個Google資助專案，目標構建一個開放、免著作權費的視訊檔格式)

- 註：src 描述video來源位址，type描述video格式(如: mp4或webm)

```
7) CrapsRules.html
1 <!DOCTYPE html>
2
3 <!-- Fig. 9.8: CrapsRules.html -->
4 <!-- Web page with a video of the basic rules for the dice game Craps. -->
5 <html>
6 <head>
7   <meta charset = "utf-8">
8   <title>Craps Rules</title>
9 </head>
10 <body>
11   <p><a href = "Craps.html">Back to Craps Game</a></p>
12   <video controls>
13     <source src = "CrapsRules.mp4" type = "video/mp4">
14     <source src = "CrapsRules.webm" type = "video/webm">
15     A player rolls two dice. Each die has six faces that contain
16     one, two, three, four, five and six spots, respectively. The
17     sum of the spots on the two upward faces is calculated. If the
18     sum is 7 or 11 on the first throw, the player wins. If the sum
19     is 2, 3 or 12 on the first throw (called "craps"), the player
20     loses (i.e., the "house" wins). If the sum is 4, 5, 6, 8, 9 or
21     10 on the first throw, that sum becomes the player "point."
22     To win, you must continue rolling the dice until you "make your
23     point" (i.e., roll your point value). You lose by rolling a 7
24     before making the point.
25   </video>
26 </body>
27 </html>
```

(video 元件)

(放多個影片連結，避免其一無法播放)



9.7 Scope Rules

Javascript 變數使用範圍

- ▶ Each **variable** (變數) in a program has a **scope** (使用範圍).
 - 註：程式中的變數有使用範圍
- ▶ **Global variables** (全域變數) or **script-level variables** (<script>內都可使用) are accessible in any part of a script and are said to have **global scope** (可全域範圍使用)
 - Thus **every function** in the script can potentially use the variables
 - 註：所有function都用存取全域變數
- ▶ **Local variable** (本地/侷部變數)
 - Declared **inside a function** have **local scope** (侷部範圍) and can be used only in that function
 - 註：本地變數只有在function內存取(意指：在function的括弧內 {...})
 - **Function parameters** (函式傳入之變數) have **function scope**
 - 註：函式傳入的參數也是侷部變數
 - If a **local variable** (侷部變數) in a function has the same name as a **global variable** (全域變數), the **global variable** is “**hidden**” from the body of the function.
 - 註：若在function內有跟全域變數相同名稱的侷部變數，則會優先使用侷部變數(全域變數會被隱藏)。

9.8 JavaScript Global Functions

Global function “全域函數”

▶ JavaScript provides some **global functions** (全域函式) as part of a **Global object** (全域物件)

- 註：JavaScript 提供多種全域函式
- 註：使用時，不必特別打上全域物件名稱，直接打方法即可。

★註：NaN (Not a Number: 非數字)

Global function	Description
<code>isFinite</code> (是否有限或非NaN是，則True; 否，則False)	Takes a numeric argument and returns true if the value of the argument is not NaN, Number.POSITIVE_INFINITY or Number.NEGATIVE_INFINITY (values that are not numbers or numbers outside the range that JavaScript supports)—otherwise, the function returns false.
<code>isNaN</code> (若非數字，則True; 否則False)	Takes a numeric argument and returns true if the value of the argument is not a number; otherwise, it returns false. The function is commonly used with the return value of <code>parseInt</code> or <code>parseFloat</code> to determine whether the result is a proper numeric value.
<code>parseFloat</code> (將字串開頭(含數字部分)轉成浮點數，成功：則回傳浮點數，失敗：則回傳NaN)	Takes a string argument and attempts to convert the <i>beginning</i> of the string into a floating-point value. If the conversion is unsuccessful, the function returns NaN; otherwise, it returns the converted value (e.g., <code>parseFloat("abc123.45")</code> returns NaN, and <code>parseFloat("123.45abc")</code> returns the value 123.45).

Global function	Description
<code>parseInt</code> (將字串開頭(含數字部分)轉成數字，成功：則回傳數字，失敗：則回傳NaN) • 支援2、8、16進位	Takes a string argument and attempts to convert the beginning of the string into an integer value. If the conversion is unsuccessful, the function returns NaN; otherwise, it returns the converted value (for example, <code>parseInt("abc123")</code> returns NaN, and <code>parseInt("123abc")</code> returns the integer value 123). This function takes an optional second argument, from 2 to 36, specifying the radix (or base) of the number. Base 2 indicates that the first argument string is in binary format, base 8 that it's in octal format and base 16 that it's in hexadecimal format. See Appendix E, for more information on binary, octal and hexadecimal numbers.

Fig. 9.10 | JavaScript global functions. (Part 2 of 2.)

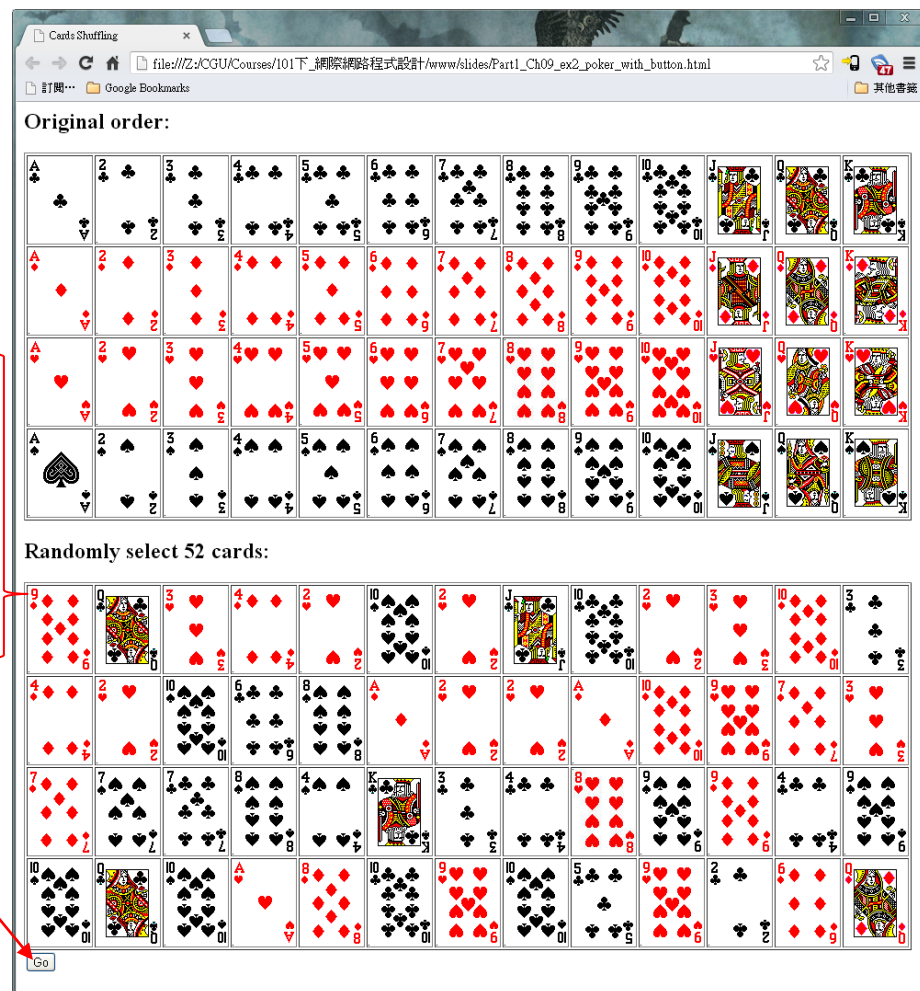


Lab

(上機練習)

In-Class Exercise #1

- ▶ 課中練習：請利用 JavaScript 產生右方表格
 - 1. 上表依序列出 52 張牌
 - 2. 當開啟網頁時，下表隨機挑選上方一張牌
 - 允許重覆挑選
 - 使用 load 事件
 - 3. 最下方加入一個按鈕來觸發重新挑選
 - 允許重覆出現
 - 使用 click 事件
 - 4. 發完牌，播放音效(win.mp3)



(圖片檔案：在ftp的.../porkers資料匣)

(Hint: 參考p12, p18範例)

In-Class Exercise #2 (Bonus / 期中小考-模擬考題之一)

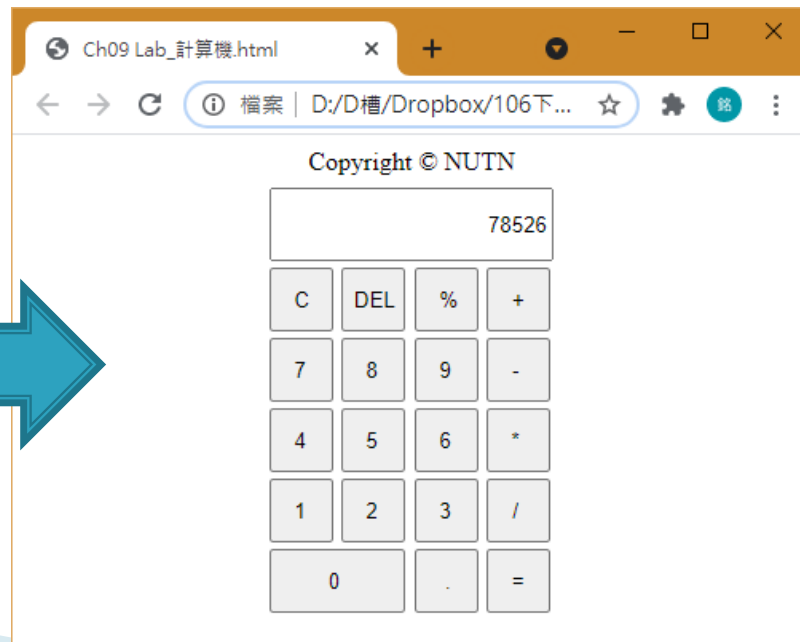
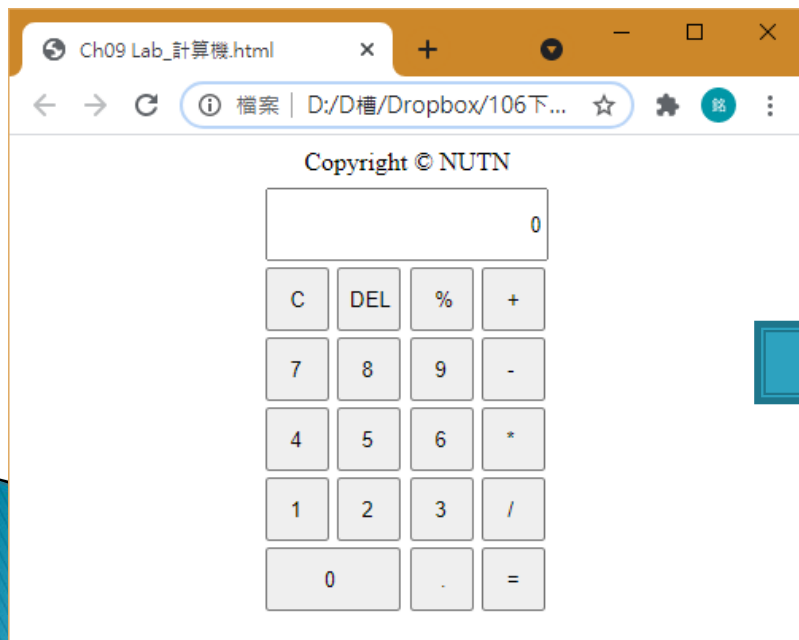
• 注意：考前猜題

▶ 課中練習：請基於html、javascript語法，建立一個外型為計算機的網頁程式，達成以下要求：

- 1. 標題為“Copyright © NUTN”
 - Hint: Ch02 (caption, special characters)
- 2. 表格包含textbox，初始值為0，數值靠右
 - Hint: Ch03(type, value, text-align)
- 3. 每個按鈕長寬大小，其中“按鈕0”為兩倍寬。
 - Hint: Ch04 (style="width: ... px; height: ... px;")
- 4. 當點擊數字鍵“0~9”則textbox會同步顯示對應數字“0~9”。
 - Hint: Ch09 (getElementById, addEventListener, .value)
- 5. 當點擊“DEL”將會移除(最右邊)一位數字
 - Hint: 補充用法 `string = string.slice(0, - 1);`

• 注意：

- 本週五(4/24) 23:59 前上傳到ftp (否則為遲交)
<ftp://111webdesign:webdesign@120.114.190.22:4000>
- Lab07
- 請以“學號”建立資料匣 (否則扣分)



課程公告：(預計) 4/28 期中小考(上機考)

Ps. 期中小考(上機考)：將提供「課程講義」及「課程範例資料」下載

月份	週次	星 期						
		日	一	二	三	四	五	六
四月	七	5	6	7	8	9	10	11
	八	12	13	14	15	16	17	18
	九	19	20	21	22	23	24	25
	十	26	27	28	29	30		
五月							1	2
	十一	3	4	5	6	7	8	9
	十二	10	11	12	13	14	15	16
	十三	17	18	19	20	21	22	23
	十四	24	25	26	27	28	29	30

本週上課 (Ch09)

期中小考 (上機考)

期中-分組報告(1)

(備用) 期中-分組報告(2)

5/5, 5/12: 期中分組報告(Group Report)



↑▼說明：進入期末專題前，讓各位同學具有分析網站優勢能力，期望能學習效法這些成功網站之元素，並提出小組期末專題構想（網站/網頁應用）。

↑▼分組報告：

★ 1) 請上網搜尋一個成功知名的網站(頁)，如：

- × 社群類網站：Facebook、Instagram、Twitter、微博、抖音 (Tictok)、小紅書、Tumblr、Pinterest, Discord, Dcard, Plurk (噗浪)等 (不限於此)
- × 電子商務平台：Amazon、淘寶、蝦皮、ebay、Yahoo拍賣、露天、比價網 等 (不限於此)
- × 募資平台、旅遊/住宿網站、論壇網站、線上直播網、串流網站、影音平台等知名網站皆可

★ 2) 請分析介紹其成功原因，報告投影片大綱應包含：

★題目越獨特，會加分

- × 1. 簡介 (歷史、組成、客戶群...)
- × 2. 功能構成元素 (3~5項核心功能)
- × 3. 成功因素 (歸納出3~5項原因、持續成功原因、和其他平台差異、解決哪些痛點)
- × 4. 天馬行空的想像 & 進化版
- × 5. 你們的期末專題提案
- × 6. 參考資料
- × 簡報範例(連結)：

https://drive.google.com/drive/folders/1QJFz8ln5BppWH_jkmWfWmHnIq0_dEqiO?usp=sharing



報告規則 & 評分標準

┆▼報告規則

- ★每組報告時間：15分鐘 (最多20分鐘，含Q&A)
- ★每組成員：1~3人一組 (最多3人)
- ★每組報告之題目(不可重覆)、順序(先報/後報)，請於期中考前4/27(一) 23:59 前mail至老師信箱: jmliang@mail.nutn.edu.tw (先寄先贏)
- ★報告順序，即時公佈在：
× https://docs.google.com/spreadsheets/d/1lCve_2rLigQKjC85pGJUttPQ3w8PAHBbBA2GGFTQrJA/edit?usp=sharing

┆▼報告評分標準：

- ★主題獨特性(25%)
- ★內容完整性(25%)
- ★提案創意性(25%)
- ★投影片設計+表達清晰流暢(25%)



▶ [工商服務]



【分享】：老師指導專題生-近期成果

- 【2025/10/23】
- 2025 NTIN-start 三創競賽
- 獲得「**第一名**」
- https://www.instagram.com/nutn_ee/p/DQZGg86knDp/





分享：老師指導專題生-近期成果

- 【2024/11/08】
- 2024全國AI創意實作競賽
- 跨域組「第一名」
- 獎金：1.5萬元
 - 連結：
 - <https://www.facebook.com/share/p/16f5A5WGT5/>





分享：老師指導專題生-前期成果

- 【2023/10/26】
- 2023智慧科技應用創意競賽
- 創新實作組「銅獎」
- 獎金：1.5萬元
 - 連結：
 - <https://www.facebook.com/100069871110907/posts/639258908413141/?mibextid=mna8qTP8xvtPE1tK>





分享：老師指導專題生-前期成果

- 【2023/11/24~26】
- 專題作品-受邀請參展於大臺南會展中心
- 「2023沙崙智慧綠能科學城」

